AI-DRIVEN PLATFORM GUIDING PADDY FARMERS ACROSS THE FULL CULTIVATION LIFECYCLE

Project Id: 25-26J-242

Project Proposal Report

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Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

AI-POWERED POST-HARVEST ADVISOR

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Dilakshan K	IT22339942	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

..... 31/08/2025.....

Signature of the supervisor: Date:

ABSTRACT

Sri Lankan paddy farmers face substantial post-harvest losses of 10-15%, with storage accounting for 4-6% of production due to high moisture content, pest infestations, and mold growth during monsoon seasons. These losses significantly impact on food security and smallholder farmer incomes.

This project proposes an AI-Powered Post-Harvest Advisor tailored to Sri Lankan conditions. The platform integrates low-cost Bluetooth sensors, smartphone camera analytics for pest identification, weather APIs, and Random Forest Regression models to predict safe storage durations for rice varieties (Bg 300, At 306). The system features offline functionality with multilingual interfaces (Sinhala/Tamil), providing actionable recommendations such as optimal storage periods and selling timing.

The advisor targets reducing storage losses to below 2% while enhancing decision-making and profitability during Maha and Yala seasons. Through pilot testing and sustainable commercialization via voluntary contributions, this solution promotes resilient farming practices using localized RRDI data for improved accuracy within Sri Lanka's agricultural sector.

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Keywords: Post-harvest losses, AI agriculture, paddy storage, pest detection, Random Forest, IoT sensors, rice varieties, smart farming

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DHT	Digital Humidity and Temperature
GPS	Global Positioning System
IoT	Internet of Things
LSTM	Long Short-Term Memory
ML	Machine Learning
RRDI	Rice Research and Development Institute
SDK	Software Development Kit
SSL	Secure Sockets Layer
SVM	Support Vector Machine
UI/UX	User Interface/User Experience

1 INTRODUCTION

1.1 Background & Literature survey

Background

The paddy sector is an important provider of food security and economic well-being in Sri Lanka with rice serving as the staple food of more than 21 million people. However, post-harvest cereal losses remain a constant problem, especially in view of the dominance of smallholder cereal producers. Durable crops like paddy are estimated to have incurred losses of 10-20% on average due to poor storage conditions, pest attacks, moisture-related damage, among other high volatility market forces. The most common storage practice is a high moisture content (above 20%), which promotes mould growth, destructive insect activity and economic pressure to sell rapidly at reduced prices [1][2].

Recent research highlights the magnitude of these problems in tropical climates like Sri Lanka where monsoon seasons increase the penetration of moisture and the spread of pests, causing damage that costs smallholders more than several million U.S. dollars annually [3]. In addition, these difficulties have been worsened by climate change, where erratic weather conditions have increased variability in harvest quality and viability during storage [4]. Smallholder farmers, who account for more than 80% of paddy producers in Sri Lanka, often have limited access to modern storage facilities or in real-time market information to make optimal decisions, thereby trapping them in poverty cycles [5].

Keeping in consideration the above identified challenges to these post-harvest management issues, the current project aims at creating an Artificial Intelligence AI-Based Post-Harvest Advisor for Sri Lankan paddy farmers. The main goal is to develop a management platform which reduces storage challenges faced by farmers in Sri Lanka. The system's ability to maximize revenue, reduce storage risks and improve decision-making capabilities is achieved through timely recommendations on optimal moisture content, pest management and market price dynamics. The technology combines the use of inexpensive sensors, smartphone-based image

processing, weather data and machine learning algorithms to give actionable advice on storage conditions, pest recognition and optimum cashing of paddy at market prices.

Using three representative cultivars (Bg 300, Bg 352, and At 306), the advisor aims to decrease storage losses from 4%-6% to below 2%, improve grain quality, and increase farmers' profitability by making informed decisions [6][7]. The study uses a Random Forest Regression model to predict safe storage period for rice varieties under different environmental, market and local socio-economic parameters. In doing so, it considers the interaction between the technologies of agriculture, economics, and context-dependent constraints to advance sustainable agroecosystems in Sri Lanka that are governed by monsoon dynamics [8].

Further complementing the accuracy is the inclusion of IoT sensors for constant monitoring that allow proactive interventions to be made in line with the global trends in smart agriculture. Sensor data can be combined with market and weather APIs to accurately predict ideal harvest times, thereby improving farmer profit [9][11].

Literature Survey

Post-harvest handling of paddy has been the subject of much research in Sri Lanka, and especially the economic and nutritional consequences have been explored. Recent literature suggests that storage losses alone have been estimated to be 4-6% of the total paddy yield, which contributes to about 15% of total post-harvest loss [1][2]. Conventional storage methods using gunny bags are susceptible to infestations by pests and moisture penetration thus increasing loss rates [3][4]. The most common pest species in Sri Lanka are the rice weevil (Sitophilus oryzae), lesser grain borer (Rhyzopertha dominica), and the Angoumois grain moth (Sitophilus oryzae) all cause both quantitative and qualitative losses through grain feeding and contamination[12][13]. Modern pest-management literature emphasizes the

importance of integrated strategies to keep pest populations at low thresholds [14][15]. Moreover, AI-enabled detection modalities, in particular convolutional neural networks for image recognition, have been shown to provide early identification of pests, thus reducing reliance on chemical control measures [16].

Seasonality also makes post-harvest decision making difficult, with rice prices varying with both the Maha (great or main, February - April) and Yala (lesser, July - August) seasons. In the post-Maha season, the prices usually fall sharply due to high supply whereas 13-16% price increase can be achieved in the intraseasonal lows [17][18]. Weak access to current market information is a major cause of farmers' suboptimal selling strategy, which is further complicated by reliance on middlemen [19][20]. Artificial intelligence-based market-prediction models that use time-series data and exogenous economic factors have the potential to reduce these volatility patterns by providing timely recommendation for farmers to sell [21].

There is potential for machine learning and artificial intelligence to improve grain storage optimization. Predictive forecasting models using sensor data, in particular temperature, humidity and CO2, have been successful for predicting storage conditions and spoilage events [22][23]. As an example, deep learning frameworks can combine multisensory measurements to detect risk in the grain storage which outperform current solutions in terms of pest and mold detection [24][25]. In addition, there have been successful applications of Random Forest models and other ML methods to predict storage life cycles using environmental variables and grain quality parameters [14]. However, studies specific to rice are few in tropical environments like Sri Lanka where most studies are for generic cereals grown in temperate climate [15].

Recent literature has found the Internet of Things an effective tool for real-time supervision; Bluetooth-enabled sensors provide data which AI algorithms can use to predict moisture-related risks [9]. Lessons learnt from other studies in Sri Lanka and the indigenous Bg 300 rice variety indicate the need for a variety specific model that

can capture the genetic diversity in storage resistance [3]. Integrating weather APIs with ML has been effective in predicting monsoon impacts, reducing mold growth by up to 30% in pilot studies [4] Additionally, the use of multilingual UIs on AI solutions has increased adoption by heterogeneous farming communities [5]. Extensive studies on smart-agriculture technologies have confirmed the potential power of AI to reduce post-harvest wastage by about 50% using predictive analytics [6]. However, the issue of data localization still looms large; data is largely derived from non-tropical latitudes and highlights the need for model adaptations for Sri Lankan conditions [7].

1.2 Research Gap

The research gap is due to the absence of integrated artificial-intelligence systems that can deliver an end-to-end post-harvest advisory service to Sri Lankan paddy farmers. Existing solutions are often not farmer-friendly and lack an integrated approach combining environmental monitoring, pest detection, quality prediction and market timing. Moreover, they disregard the regional issues such as seasonality of monsoon, diversity of rice varieties and limited connectivity.

In order to overcome such post-harvest issues, the proposed system uses a safe storage prediction based on real-time sensor information and weather forecasts. It uses smartphone cameras and low-cost instrumentation to enable early detection of storage pests and mold risks. Moreover, the platform also incorporates seasonal market price trends to suggest bestselling times and accounts for moisture and temperature differences due to monsoon rains that Sri Lanka is unique for. Specific support for the most common rice varieties of Sri Lanka is given by employing variety-specific models, and multilingual support in Sinhala and Tamil was provided to optimally target rural farmers. The integrated approach alleviates the disjointed nature of existing tools, which are often focused on single aspects of pest detection, without some form of economic or climatic information specific to the region.

.

Table 1-Comparison table

Research	Storage Duration Prediction	Pest Detection	Market Price Integration	Multilingual Support	Sri Lanka Specific Focus
Using machine learning to estimate and forecast the effect of storage and packaging on paddy rice quality	✓	✗	✗	✗	✗
Enhancement of machine learning models to forecast the alteration of glutinous rice storage quality at high temperature	✓	✗	✗	✗	✗
Monitoring and forecasting corn grain quality during transportation and post-harvest activities in the storage facilities by using sensors and machine learning	✓	✓	✗	✗	✗
Applications of Machine Learning in the Horticultural Industry and Future of Postharvest Quality	✓	✗	✗	✗	✗
Suggested Artificial Intelligence-Based Post Harvest Advisor	✓	✓	✓	✓	✓

1.3 Research Problem

Paddy is the mainstay of national food security and rural livelihoods in Sri Lanka, yet the sector faces serious threats from high post-harvest losses that erode

productivity and incomes, especially for smallholder farmers who make up the majority of producers in the sector.

Estimates suggest that post-harvest losses for hardy crops like paddy fall in the range of 10 - 20 %, while earlier estimates have been as high as 25 - 30 %. These losses are largely due to poor storage practices, high moisture content at harvest (often >20%), attack by rice weevil (Sitophilus oryzae) and lesser grain borers (Rhyzopertha dominica), mold growth during monsoon periods and poor market timing that forces premature sales at low prices. The subsequent losses are aggravated by the continued operation of the conventional storage systems (gunny bag storage) which lead to an overall grain quantity and quality losses of 15-20% over the storage period through the post-production operations. Further, just the harvesting losses are estimated to account for around 6% of total production and extra economic costs due to transportation inefficiencies, if ameliorated appropriately, could result in savings in the order of several million rupees.

The lack of official government statistics on paddy and rice stocks also complicates the proper estimation of losses and policy formulation, setting in motion a cycle of food insecurity, depressed farmer incomes, and strain on the national economy, in a country where rice formed the staple of the food diet for millions. Smallholders, often working in remote areas ill served with real-time information about weather, markets, or pest risks, engage in crude methods that do not mitigate the vulnerabilities to tropical climates and produce unnecessary waste that undermines sustainable agriculture and increases poverty.

This complex issue has wide-ranging implications not only for short-term yields but also for the long-term viability of agriculture, as erratic weather patterns become more pronounced due to climate change. Thus, localized technological interventions are urgently needed to reduce post-harvest losses and improve decision making in post-harvest management.

2 OBJECTIVES

2.1 Main Objectives

The overall goal of this study is to propose an AI-Powered Post-Harvest Advisor tool to help reduce losses when storing paddy and increase farmers profits in Sri Lanka. This would require a combination of cheap sensors, machine learning models and market data to offer practical guidance on how and when to store rice; detect pests or bugs; and how and when to sell, adjusted to local varieties of rice and seasonal issues.

2.2 Specific Objectives

Develop a data collection framework

Collect environmental, storage and market data which is highly specific to paddy farming in Sri Lanka including sensor-based measurements of temperature and humidity, pictures of pests and past price trend of Maha and Yala seasons.

Build and train machine learning models

Utilize Random Forest Regression to forecast safe periods of storing rice according to the moisture contents, rice variety (Bg 300, At 306), and other weather conditions to determine the model that is trained on localized data labelled as the preferred equipment and storage capacity.

Integrate detection and monitoring tool

Combine with the ability to conduct a smartphone camera analysis and Bluetooth sensors to monitor those pests and mold in real-time, alerting farmers about such risks as rice weevil infestations or moisture spoilage.

Create user-friendly advisory interface

Design a mobile app with simple interfaces (Sinhala and Tamil) that will recommend storage periods (e.g. store this another 2 week or sell it) by combining assessments on the availability of storage facilities and fluctuations in market rates.

Evaluate and refine the system

Conduct pilot tests with farmers to assess losses reduction (of less than 2%) and to collect feedback on system usability and iterate systems to meet its performance and ethical data management needs.

3 METHODOLOGY

3.1 System Architecture

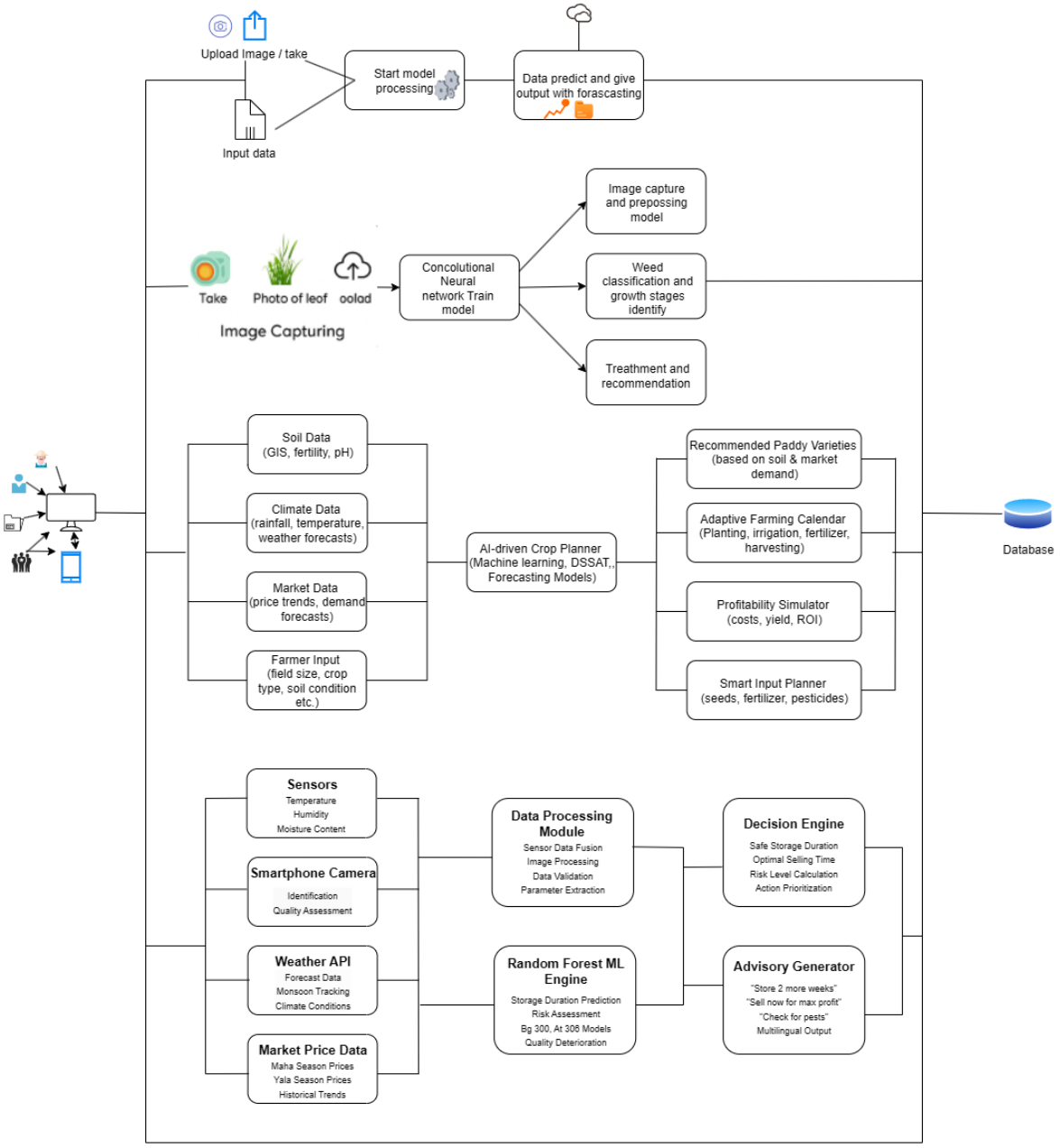


Figure 1:System Architecture diagram

The diagram illustrates an integrated AI-based platform to paddy growth in a flowchart that divides the system into four related parts, each of which corresponds

to the duties of a team member, to guide farmers through the entire cultivation cycle. Starting with the sources of input data (sensor readings (temperature, humidity, moisture), smartphone images, weather APIs, market price datasets, soil/GIS data, climate forecasts, and farmer-provided data (field area, crop type, soil condition) the system combines this multimodal information through data processing modules, which include modular data fusion, sensor integration, routine validation and parameter extraction. The centralized processing simplifies these outputs to machine learning models such as convolutional neural network models to interpret images, random forest classifiers to provide predictive analytics, and AI models to plan. The recommendations that follow are stored in a database to inform adaptive farming schedules, profitability models, intelligent input schedules, weed sorting, intervention advice, and post-harvest advice.

The segments integrated are: first, Pest Attack Forecasting which predicts possible attacks with alerts using historical data and real time data via multimodal inputs with dynamic learning; second, AI-Powered Weed and Damage Detection Stages Identification which uses computer vision to classify and assess damage to weeds and recommend treatments which are staged and integrated with pest data for comprehensive management of the fields; third, Crop Establishment Planner for New Farmers which recommends varieties along with soil and climate inputs, designing the calendar as well as estimating profit to aid foundational plan which shapes monitoring, and finally, AI-Powered Post-Harvest Advisor which uses classifying, weather information, and market data to optimally time storage and selling, completing the cycle by utilizing data from previous steps to reduce wastage, ensuring precision, sustainability, and data in all steps.

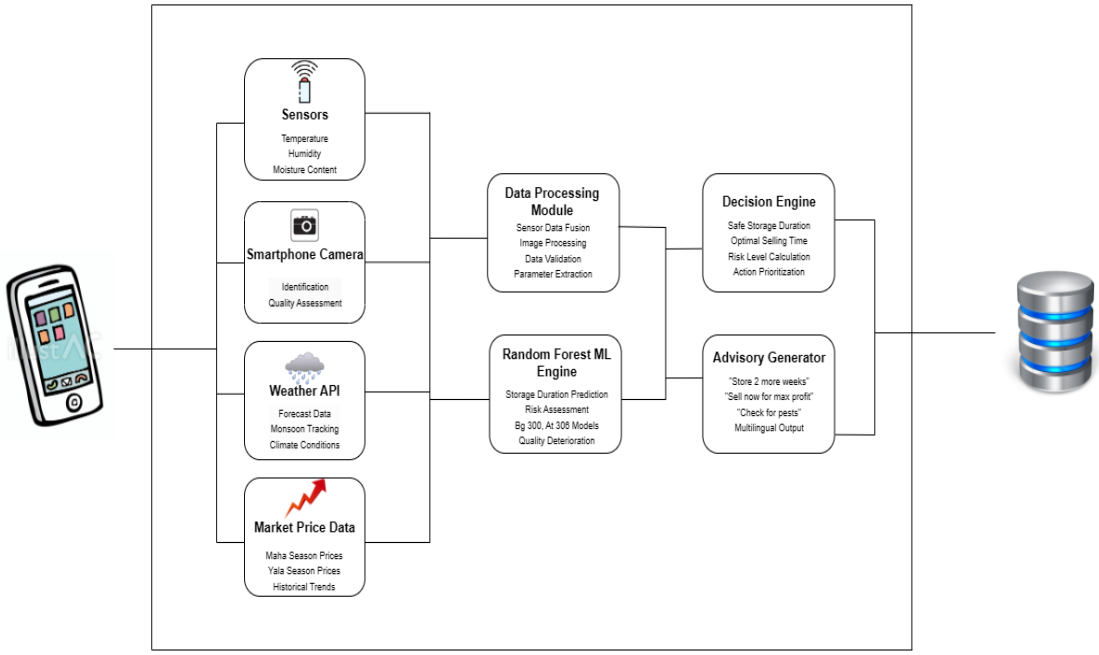


Figure 2:Component diagram

This diagram depicts the AI-Powered Post-Harvest Advisor as a subsystem of the paddy farming platform, which is to support optimization of storage and marketing decisions through the combination of real-time and historical data streams, reducing loss and increasing profitability. Environmental sensor (temperature, humidity, moisture content), smartphone camera (used to check quality) data, weather information services (forecast data, monsoon tracking, climatic conditions), market price repositories (seasonal, historical price trends) data is fed into a data-processing module. In this module, sensor fusion, image analysis, data validation, and parameter extraction are conducted to make ready clean multimodal inputs. Such polished datasets, in turn, drive a Random Forest machine-learning engine, which predicts the best storage times, determines risk based on specialized models, and identifies quality decay, thus allowing the decision engine to calculate safe storage periods, ideal seller-time windows, risk, and prioritized actions. The advisory generator then generates actionable information that is stored in a database, such as storage advice, selling advice, pest-inspection advice, and multi-lingual notifications, thus ensuring that farmers get evidence-based and customized advice regarding how to deal with post-harvest problems. This cross-linked workflow

fosters accuracy in the post-harvest management through AI-driven analysis of the dynamics of environmental, quality, and market variables, thus completing the cultivation cycle with either sustainable or economically feasible results.

Technologies	Python, React Native, Flask Server, TensorFlow, OpenCV, VS code, Firebase
Techniques	Machine Learning, IoT Integration, Data Analytics, Image Processing
Algorithms	Random Forest Regression, CNN, K-Means Clustering, SVM, Decision Trees
Architectures	MobileNet V2, ResNet-50, YOLO v5, Microservices Architecture, RESTful APIs

3.1.1 Software Solution

- Requirement gathering

  - Data gathering

The Bathalagoda Rice Research Institute was used as the source of data, including the datasets of the post-harvest parameters of the rice varieties including Bg300 and At306. The data was 500 observations on moisture threshold, the presence of pests and the expansion of models based on experimental farms, as well as seasonal reports about yields. Before model training, the data were anonymized to enhance predictive accuracy under Sri Lankan agronomic conditions.

  - Conducting a survey

The structured questionnaire was used to conduct a survey to interrogate post-harvest challenges, advisory preferences, and attitudes toward the adoption of artificial intelligence. The resulting reactions helped to clarify the need of interventions to reduce losses, offline availability, and multilingual guidance, thus informing the future refinements.

Sequence Diagram

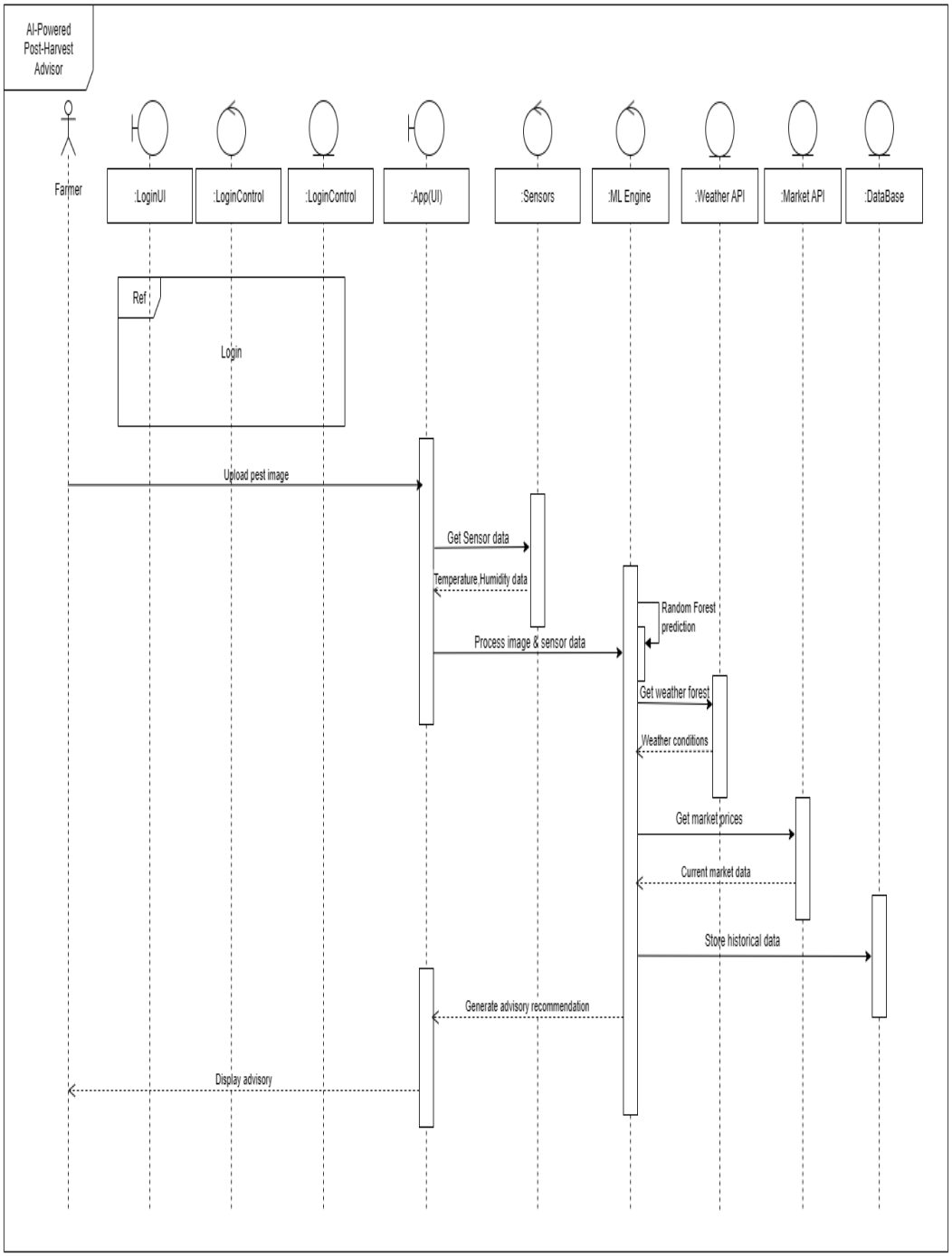


Figure 3:sequence diagram of Ai-powered post harves advisor

Use Case Diagram

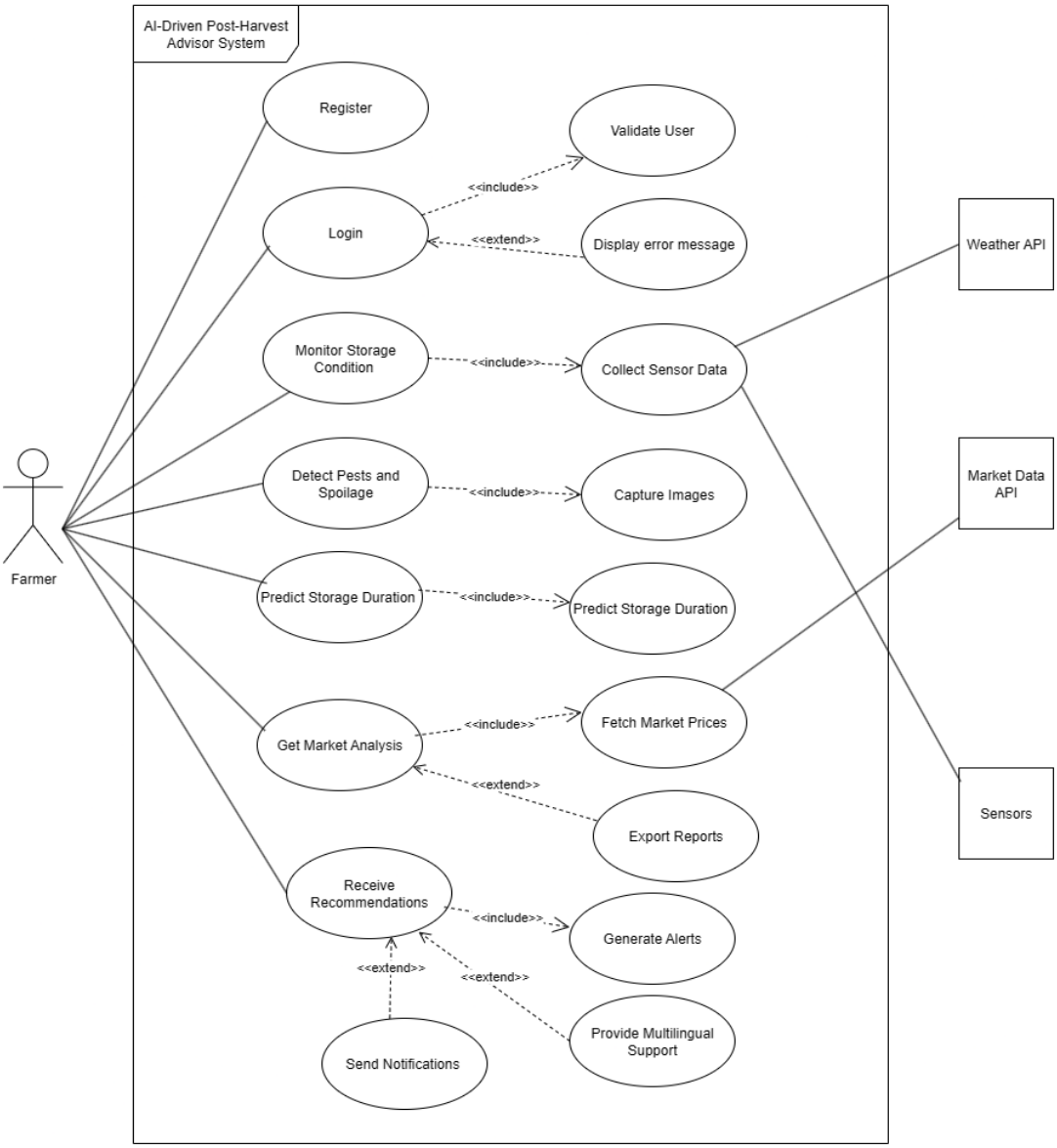


Figure 4::Use case diagram of AI powered post harvest advisor system

- **Implementation (Development)**
  - **Implementation of cross-platform mobile application using React Native**

    The interface supports both the Sinhala and Tamil languages.

    The architecture incorporates pest detection imaging with cameras.

  - **Sensor Data Collection System Development**

    Integrated with Bluetooth DHT22 temperature - humidity sensor.

    Real-time data streaming is deployed along with strict data integrity validation mechanisms.
- **Testing**
  - **Unit Testing**

    Unit tests - individual components are tested for correctness (machine-learning models, API endpoints, sensor-data processing agents, etc.) when run in isolation.
  - **Integration Testing**

    A coherent communication between the mobile application and backend services and the Internet of Things sensors is evaluated to ensure that such subsystems seamlessly communicate without any degradation of the functions.
  - **System Testing**

    An end-to-end analysis of all the workflows from the acquisition of the data to the generation of the advisory is conducted to verify system integrity and functionality.
  - **User Acceptance Testing**

    For this, field-based evaluation is conducted with farmers to validate the utility and effectiveness of the recommendations offered by the system
  - **Accuracy Testing**

    The machine-learned outputs are used to compare with storage reality and observed pest-detection results to validate the accuracy and reliability of the system's recommendations.

3.1.2 Commercialization

The Post-Harvest Advisor powered by AI leaves much room to sustainable commercialization in Sri Lanka. The app also can directly support the increase of farmers incomes by assisting farmers to minimize post-harvest waste, preserve food stores by using best methods, and get information about the market to guide farmers in the sale of their products. In its turn, it reinforces the agricultural sector in general and has a positive effect on the national economy.

The system is economically efficient to guarantee that it is affordable to even the smallholders in the rural locales. The peculiar feature of the commercialization model is the approach to its financing The financing of the app comes in the form of a voluntary contribution by the farms that already benefit: when they enjoy the app, and are willing to donate a small amount of money to help maintain and expand the app. By giving these donations, farmers do not perceive it as a burden but as a way of helping other farmers who are yet to make profits.

The model will provide a self-sustainable ecosystem in which not only farmers will be empowered but also the app will be used as the continuous revenue generation source that will host the long-term operation. The tool can allow a balance in the social and economic viability at the same time, aiding the expansion of commercialization of agriculture sector in Sri Lanka.

3.1.3 Future Scope

- Multi-Crop Compatibility: Extend the advisor to additional staples, such as maize or vegetables, coming up with variety-specific models across different agro-ecosystems of Sri Lanka.
- AI-Driven Education: Embed training on sustainable techniques including hermetic storage, to create capacity among farmers and lower their dependency on chemical control measures.
- Advanced Monitoring: Introduce drone imaging or wearable devices to overarching farms to optimize the precision of pest detection and storage.

4 PROJECT REQUIREMENTS

4.1 Functional requirements

- User Inputs Analysis: Process sensor data, images, and manual entries in order to assess accurate storage conditions.
- Real-Time Condition Monitoring: Monitor temperature, humidity and pest indicators continuously
- Dynamic Response Adaption: Provide custom recommendations (e.g. aerate, sell in 10 days) based on the ML predictions and market trend.
- Dialog Management: Provide users with natural language answers, engaging in multiple language interactions to embrace accessibility.
- Privacy and Data Protection: Agree environmental laws by using consent-based data collection and encryption to meet local laws.
- Enhance the user experience: Offer user-intuitive UI with visualization such as storage risk charts.
- User feedback and Rating system: Get in-app reviews to develop features further.
- APIs and Integrations: Connect to weather and market APIs for real-time data fetching. Connect to weather and market APIs for real-time data fetching.

4.2 Non-functional requirements

- Performance: Best predictions should occur in less than 5 secs on low end phones.
- Accuracy and Personalization: Achieves >85% accuracy in storage forecasts which is customized to user-specific varieties and locations.

- Usability and User Experience: Easy, graphic interface to accommodate low-tech users.
- Reliability and Availability: Offline capability and 99 percent uptime when online.
- Security: Data encryption and secure APIs to prevent breaches.
- Compatibility: Support Android 8+ and iOS 12+, common in Sri Lanka.

4.3 System requirements

- **Software Requirements:**
  - Development tools (VS Code, Android Studio, Python, Node.js)
  - Machine learning stack (TensorFlow, OpenCV, Scikit-learn)
  - Mobile development framework (React Native with necessary libraries)
  - Database and API integration requirements
- **Hardware Requirements:**
  - Mobile device specifications (Android/iOS with camera and Bluetooth)
  - IoT sensor requirements (DHT22, ESP32, power systems)

4.4 User Requirements

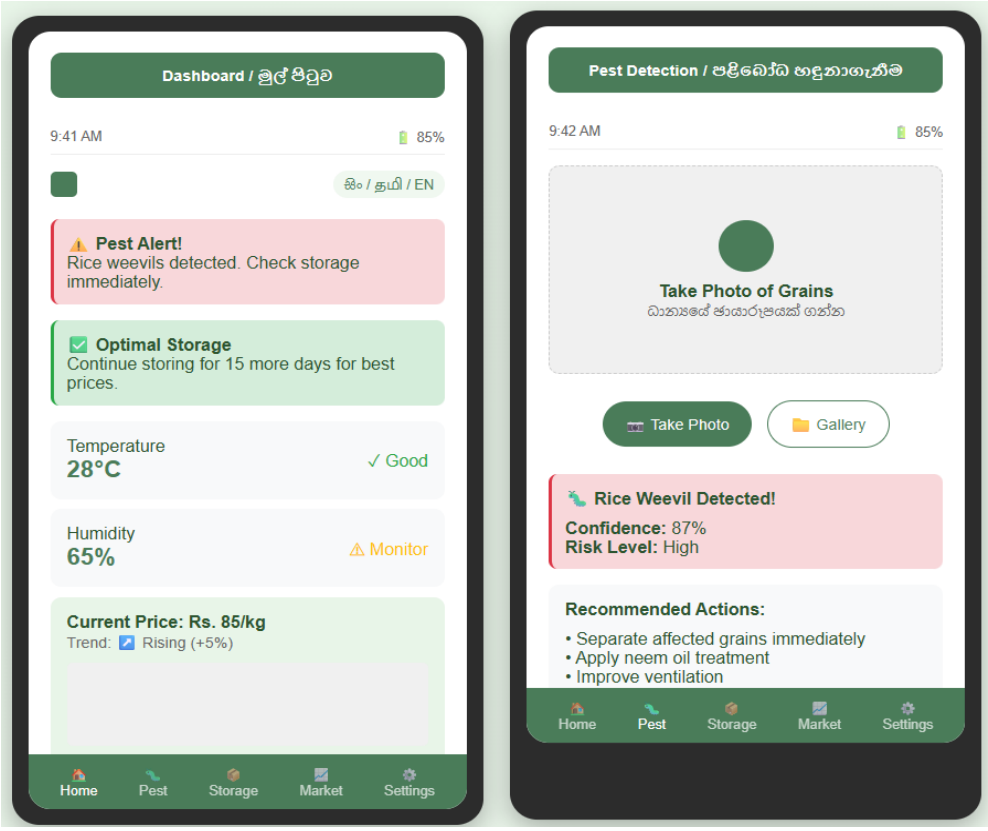
This section outlines the key user requirements of the AI-Driven Post-Harvest Advisor platform with a sense of the end-user requirements in the Sri Lankan paddy farming environment. The requirements highlight feasible functions including real-time monitoring, pest detection, storage prediction and decision-making support to reduce losses during post-harvest.

1. **Real-Time Environmental Monitoring**: The users must be in a position to draw interconnection between cheap Bluetooth sensors and a mobile app to constantly measure the temperature and humidity of storage facilities. The system should provide prompt warnings when these parameters exceed set safe values, thus

enabling proactive measures to be taken- like aeration- to prevent the growth of mold and spoilage of products during monsoon seasons.

2. **Pest and Spoilage Detection**: Through smartphone camera, users will be in a position to post images of stored paddy to be analyzed by AI. The site should be able to identify pests and mold threats precisely and provide real-time rectifying advice in multiple languages to minimize quality losses and damages by 4-6 per cent to less than 2 per cent.
3. **Safe Storage Duration Prediction**: The user should provide data on rice-variety, environmental conditions, and current conditions; the system then uses the Random Forest regression approach to approximate the best storage times.
4. **Market Price Analysis and Selling Advice**: Integrated climatological forecasts and real-time market price datasets will be provided to stakeholders by accessing the data of agencies like RRDI, thus allowing the provision of individual selling recommendations. The application, which will have the ability to be used fully offline, will support scenario-simulation analysis to assess profitability, thus optimizing the decision-making process and eventually maximizing the income of an industry majorly consisting of smallholder producers.

4.5 Wireframes



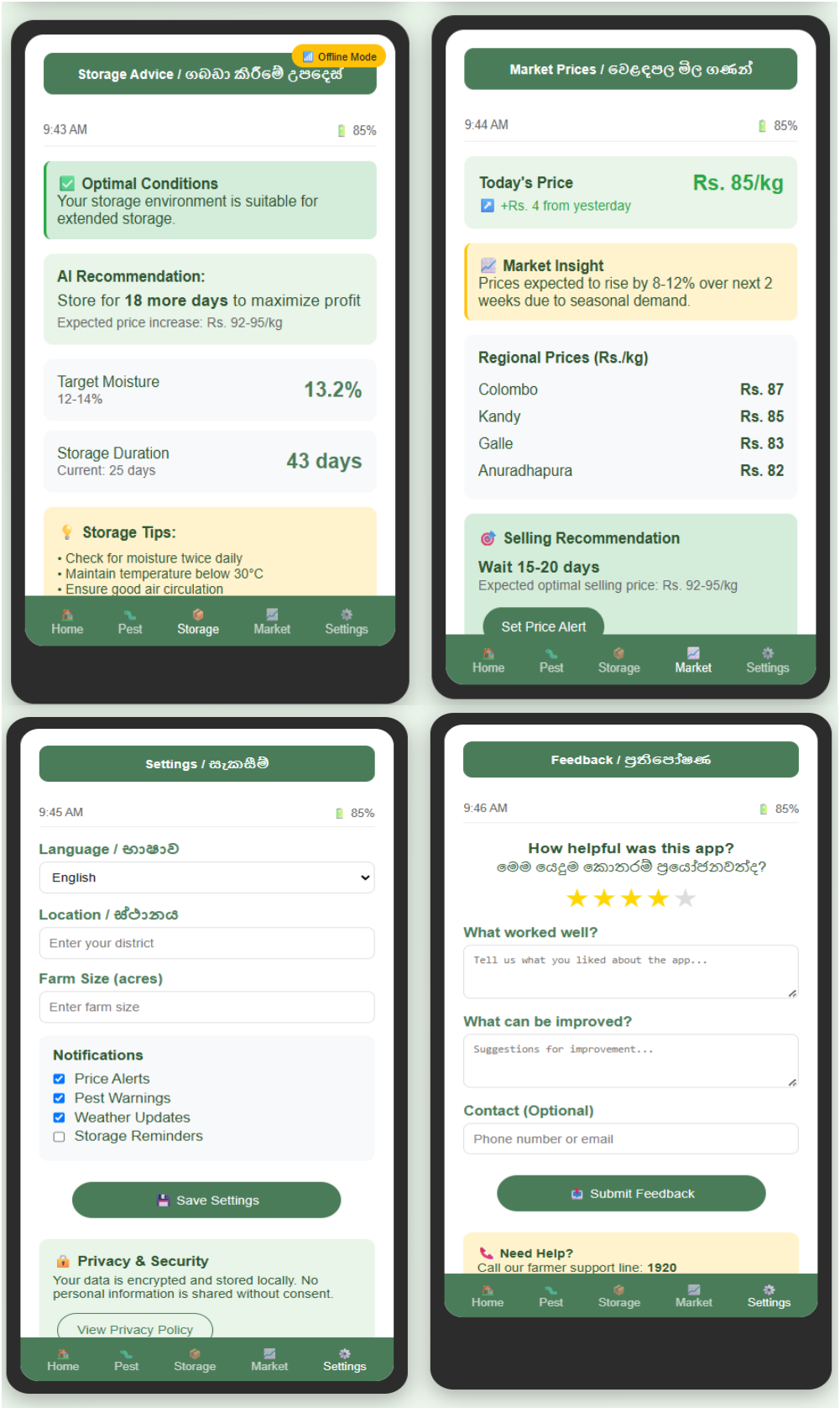


Figure 5:wireframes

5 GANTT CHART

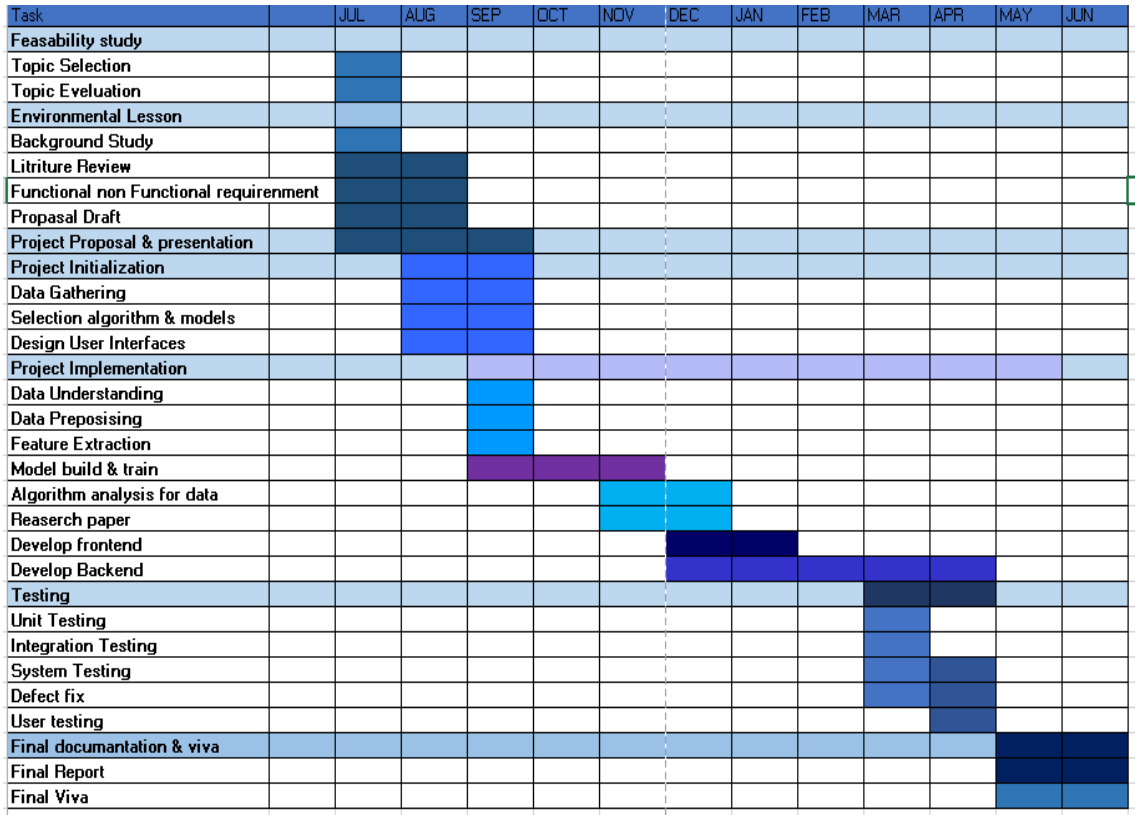


Figure 6:GanttChart

6 WORK BREAKDOWN CHART

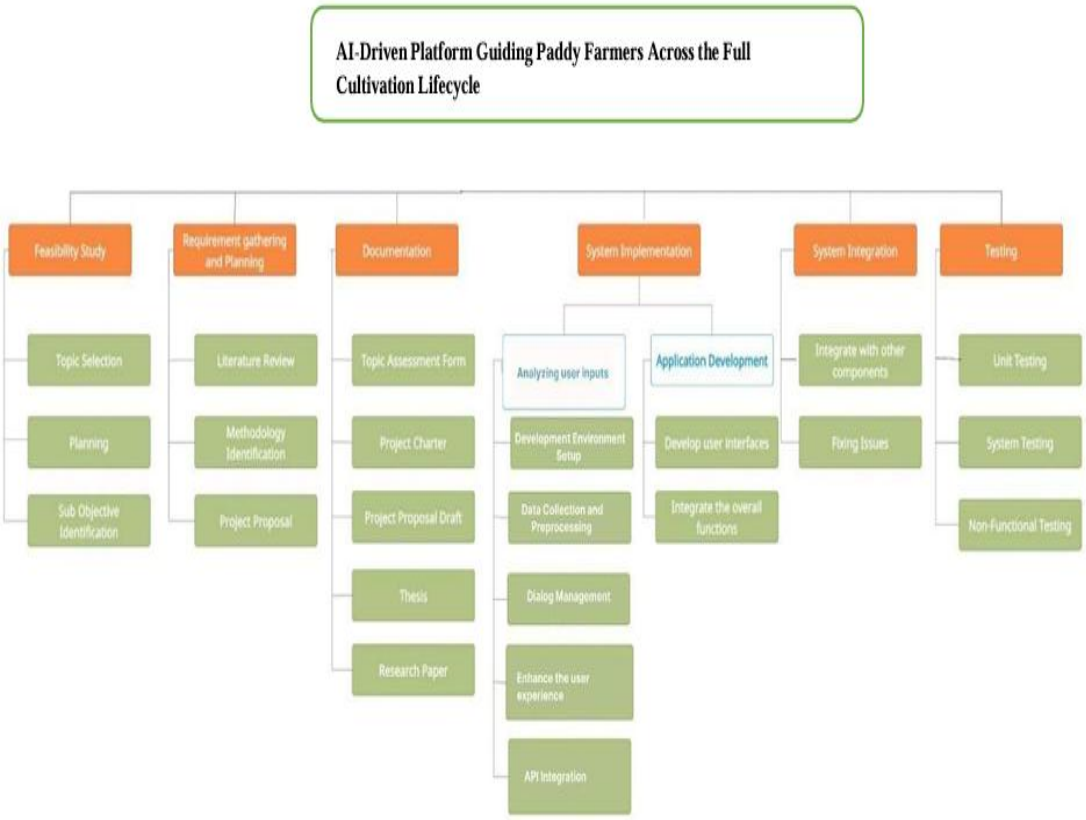


Figure 7:WorkBreakDownChart

7 DESCRIPTION OF PERSONAL AND FACILITIES

Supervisor

Ms. Lokesha Weerasinghe

Senior Lecturer

Faculty of Computing | Information Technology

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Co-Supervisor

Ms. Shalini Rupasinghe

Assistant Lecturer

Faculty of Computing | Software Engineering

Sri Lanka Institute of Information Technology | Malabe

External Supervisor

Mrs. Nandani Rupasinghe

Agriculture research and Production Assistant

Department of Agrarian Development | Sri Lanka

8 BUDGET AND BUDGET JUSTIFICATION

For this research component, we do not aim to commercialize our individual components and the whole research will be commercialized as an application. Following is the budget justification for the whole research system.

Table 1 --- Budget Table

Description	Cost(LKR)
Development tools and licenses	10,000.00
Backend Hosting	12,000.00
Cloud computing	15,000.00
Field surveys & travel	20,000.00
Paper Publishing	15000.00
DHT22 Temperature Humidity Sensor	3000.00
Total	**75,000.00**

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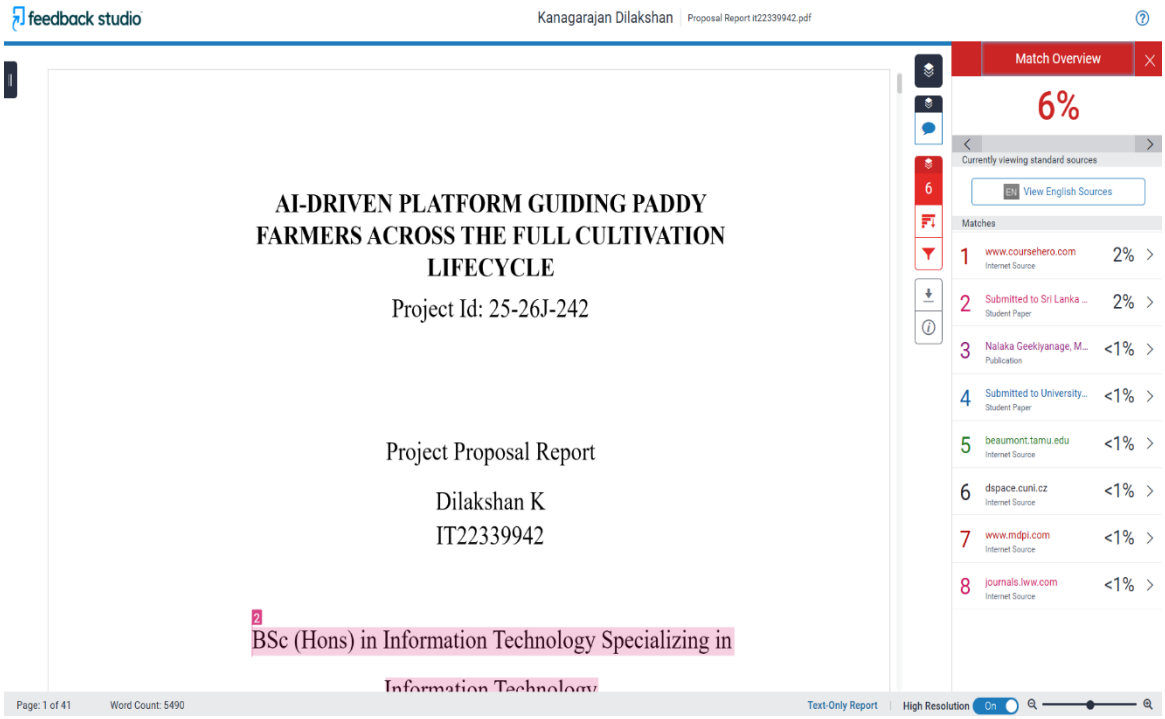
10 APPENDICES

Appendix - A: Plagiarism Report



Appendix 1 -- Plagiarism Report